COGNISENSE & COGNITWIN : END-TO-END SYSTEM ARCHITECTURE

Multimodal Continuous Patient Digital Twin for Early Alzheimer's & MCI Screening | Screening aid, not a diagnosis

TIER 1: MULTIMODAL INGESTION & SENSING LAYER

Passive Smartwatch Telemetry
CONTINUOUS (7-DAY)

- Continuous Heart Rate / RMSSD HRV
- Sleep Architecture (Deep, REM, Fragmentation)
- Daily Steps, Active Minutes & Gait

Active 8-Min Cognitive Tasks
ON-DEMAND CLINICAL

- Picture Description (Cookie Theft scene)
- Immediate & Delayed Word Recall (10 words)
- Digital Motor Trail-Making (Trails A & B)

Static EHR & Laboratory Baselines
LONGITUDINAL EHR

- Demographics: Age, Education, Sex
- Clinical Baselines: Baseline MoCA (10-30)
- Labs: Vit B12, TSH, Chol, ApoE-e4, Comorbidities

TIER 2: SECURE EDGE INGESTION & IN-CLINIC MICROSERVICES

Local Privacy Audio Pipeline
LOCAL ASR (<3s)

- faster-whisper local on-device ASR
- Acoustic pause & hesitation duration tracking
- Zero cloud audio transmission (Ephemeral delete)

Ingestion Quality Control & De-ID
DATA INTEGRITY

- Hardware & Audio SNR Quality Control Gate
- Wearable off-wrist dropout detection (<0.65)
- Presidio PII Transcript Anonymization

FastAPI REST Server & Session Store
FASTAPI MICROSERVICE

- High-throughput async endpoints (/v1/twin, /screen)
- Thread-safe SQLite persistent store (data/cognitwin.db)
- Session proxies surviving multi-worker restarts

TIER 3: COGNITWIN CORE INTELLIGENCE & ANALYTICS ENGINE

Daily Patient State Engine
STATE ENGINE

- Personal baseline calibration (Days 1-3)
- Deterministic state machine (5 clinical states)
- Temporal strictness (Zero future data leakage)

Calibrated 30-Day Predictive Model
0.982 ROC-AUC

- Grouped 5-Fold Cross Validation (by subject)
- Platt Sigmoid Calibration (0.044 Brier score)
- Wilson 95% Confidence Intervals (4 Risk Tiers)

XAI & What-If Counterfactual Sim
TREESHAP & SIMULATOR

- TreeSHAP feature attributions vs. cohort median
- Plain-language direction-aware clinical narratives
- Real-time slider intervention sensitivity simulation

TIER 4: CLINICAL PRESENTATION & HEALTHCARE INTEROPERABILITY

Clinician Digital Twin OPD
STREAMLIT 1.31

- Interactive Day 1 -> 7+ timeline scrubber
- State comparison (Day A vs Day B deltas)
- Real-time What-If scenario manipulation

Patient & Senior Web Portal
REACT 18 + VITE

- Senior-accessible UI (Atkinson Hyperlegible)
- Multilingual: English, Hindi, Marathi
- Zero-PHI local browser state machine

EHR Export & Referral Network
HL7 FHIR R4

- HL7 FHIR R4 RiskAssessment & Observation Bundle
- GDPR Art 17 Cryptographic Right-to-Erasure
- Pune Nearest Memory Clinic Locator (Leaflet maps)

Zero Cloud Audio Transmission • Subject-Grouped Zero Leakage CV • Platt Calibrated Probabilities • HL7 FHIR R4 Interoperable